EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction; Thermodynamic terms; Types of the system

1. A system in which exchange of energy is allowed but not matter:
(A) open system (B) closed system
(C) isolated system (D) all of the above

CTF077

2. An example of closed system is :
(A) Hot water present in hot beaker
(B) Some amount of water present in equilibrium with it's vapour in a closed and insulated container
(C) Some amount of hot water enclosed in a closed container which is not insulated
(D) none of these

CTF078

3. Since exchange of energy & matter is not allowed in isolated system :
(A) system & surrounding are always separated by only real boundary.
(B) system & surrounding are always separated by only imaginary boundary.
(C) system & surrounding are always separated by a real or imaginary boundary.
(D) none of these

CTF079

Raiker's Academy JEE

Extensive and Intensive properties

4. If x and y are arbitrary extensive variable then,
(A) (x + y) is an intensive variable (B) $\frac{x}{y}$ is intensive variable
(C) (x – y) is intensive variable (D) x × y is intensive variable

CTF080

Reversible and Irreversible process

5. What is true for a cyclic process [E = internal energy]
(A) w = 0 (B) $\Delta E = 0$ (C) $\Delta H = 0$ (D) B & C both

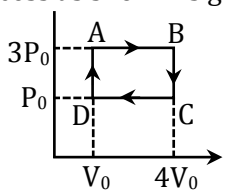
CTF087

Work Done

6. The work done during the expansion of a gas from a volume of 4 dm^3 to 6 dm^3 against a constant external pressure of 3 atm is -
(A) –608 J (B) +304 J (C) –304 J (D) –6 J

CTF091

7. What is the work done in cyclic process as shown is given P–V diagram?

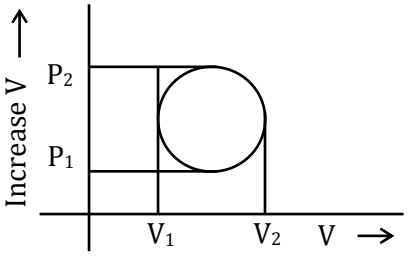


(A) $-6P_0V_0$ (B) $-12P_0V_0$ (C) $9P_0V_0$ (D) $-P_0V_0$

CTF094

For More Join: @IITJEE_Advanced

8. In the cyclic process in P-V diagram, the magnitude of the work done is



(A) $\pi\left(\frac{P_1 - P_2}{2}\right)^2$ (B) $\pi\left(\frac{V_2 - V_1}{2}\right)^2$

(C) $\frac{\pi}{4}(P_2 - P_1)(V_2 - V_1)$ (D) $\pi(V_2 - V_1)^2$

CTF101

Internal Energy

9. For and ideal gas $\frac{C_P}{C_V} = \gamma$ the molecular mass is M_w, its specific heat capacity at constant volume is :

(A) $\frac{R}{M(\gamma - 1)}$ (B) $\frac{M}{\gamma - 1}$ (C) $\frac{\gamma RM}{\gamma - 1}$ (D) $\frac{\gamma R}{M(\gamma - 1)}$

CTF109

Enthalpy

10. Enthalpy of the system is given as :-

(A) H + PV (B) U + PV (C) U – PV (D) H – PV

CTF108

Rankers Academy JEE

First Law of Thermodynamics

11. The internal energy change when a system goes from state A to B is 40 kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy?

(A) < 40 kJ (B) Zero (C) 40 kJ (D) > 40 kJ

CTF088

12. One mole of a gas absorbs 200J of heat at constant volume. Its temperature rises from 298 K to 308 K. The change in internal energy is :-

(A) 200 J (B) –200 J (C) $200 \times \frac{308}{298}$ J (D) $200 \times \frac{298}{308}$ J

CTF089

13. If work done by the system is 300 joule when 100 cal heat is supplied to it. The change in internal energy during the process is :

(A) – 200 Joule (B) 400 Joule (C) 720 Joule (D) 120 Joule

CTF097

14. The work done by a system is 8 J when 40 J heat is supplied to it. The change in internal energy of the system during the process :

(A) 32 J (B) 40 J (C) 48 J (D) –32 J

CTF098

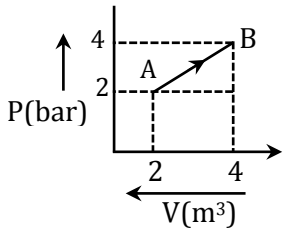
For More Join: @IITJEE_Advanced

15. A system undergoes in which $\Delta E = +300$ J while absorbing 400 J of heat undergoing an expansion against 0.5 bar. What is the change in the volume (in L).

(A) 4 (B) 5 (C) 2 (D) 3

CTF099

16. 1 mole of gas perform the following operation from A to B



If 500 KJ heat is absorbed in this process the ΔU from A to B is :

(A) –100KJ (B) +200KJ (C) +300KJ (D) –400KJ

CTF100

Degree of freedom

17. Which has maximum internal energy at 298 K?

(A) helium gas (B) oxygen gas (C) ozone gas (D) equal

CTF092

Poisson's ratio

18. For a monatomic gas, the value of the ratio of $C_{p.m}$ and $C_{v.m}$ is :-

(A) $\frac{5}{3}$ (B) $\frac{7}{5}$ (C) $\frac{9}{7}$ (D) $\frac{9}{11}$

Rankers Academy JEE

CTF107

Isothermal Process

19. An ideal gas is expanded isothermally from 3L to 5L volume at constant pressure of 1.5 atm. What is the change in internal energy of system (in atm L) ?

(A) +3 (B) –3 (C) –6 (D) Zero

CTF095

20. Maximum work can a gas do, if its alone of expand isothermally against?

(A) vacuum (B) high pressure of surrounding

(C) low pressure of surrounding (D) atmospheric pressure

CTF093

21. Work done in isothermal reversible process for ideal gas is-

(A) $-nRT \ln\left(\frac{V_2}{V_1}\right)$ (B) $-P(V_2 - V_1)$ (C) $\left(\frac{P_2V_2 - P_1V_1}{\gamma - 1}\right)$ (D) 0

CTF117

Isochoric process

22. Which of the following is/are correct for Isochoric heating of an ideal gas :

(A) $\Delta T = 0$ (B) $\Delta U = 0$ (C) $q = 0$ (D) $w = 0$

CTF119

For More Join: @IITJEE_Advanced

Free expansion

23. Two moles of an ideal gas expand spontaneously into vacuum. The work done is :–

(A) Zero (B) 2 J (C) 4 J (D) 8 J

CTF090

Adiabatic process

24. Two moles of an ideal gas ($C_V = \frac{5}{2}R$) was compressed adiabatically against constant pressure of 2 atm, which was initially at 350 K and 1 atm pressure. The work involve in the process is equal to -

(A) 250 R (B) 300 R (C) 400 R (D) 500 R

CTF121

Comparison between Isothermal and Adiabatic process

25. The slopes of isothermal and adiabatic curves are related as-

(A) Isothermal curve slope = adiabatic curve slope

(B) Isothermal curve slope = $\gamma \times$ adiabatic curve slope

(C) Adiabatic curve slope = $\gamma \times$ isothermal curve slope

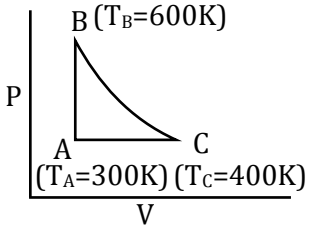
(D) Adiabatic curve slope = $\frac{1}{2} \times$ isothermal curve slope

Rankers Academy JEE

CTF120

Polytropic process

26. A heat engine carries one mole of an ideal diatomic gas around the cycle as show in the figure, the amount of heat added in the process AB & heat removed in the process CA are :



(A) q_{AB} =750R & q_{CA} =–450R (B) q_{AB} =650R & q_{CA} =–350R

(C) q_{AB} =750R & q_{CA} =–350R (D) q_{AB} =650R & q_{CA} =–450R

CTF110

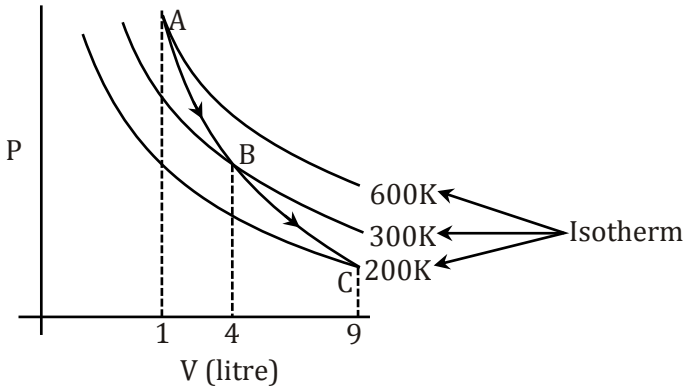
27. 2 moles of an ideal monoatomic gas undergoes reversible expansion from (4 L, 400 K) to 8L such that $TV^2 = k$. The change in Enthalpy of the gas

(A) –1500 R (B) – 300 R (C) 1500 R (D) 3000 R

CTF111

For More Join: @IITJEE_Advanced

28. The given figure shows a polytropic process ABC for one mole of an ideal gas. Calculate the polytropic index (x) for the process-



(A) ½ (B) –1/2 (C) 3/2 (D) None

CTF123

29. A gas expands against a variable pressure given by $P=\frac{10}{V}$ atm, where V is volume of gas of each stage of expansion. Further during expansion from volume 10 L to 100 L, the gas undergoes a change in internal energy of 420 J. How much heat is absorbed by the gas during expansion?

(A) q = 2752.2 J (B) q = 2652.2 J (C) q = 2852.2 J (D) q = 2452.2 J

CTF124

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Introduction; Thermodynamic terms; Types of the system

1. For an isolated system the wall/boundary separating the system from surrounding must be :

(A) rigid
(B) adiabatic
(C) impermeable
(D) diathermal

CTF081

The State of the system; State function; Path function

2. Which of the following is/are state function ?

(A) Enthalpy
(B) Heat
(C) Entropy
(D) Gibb's force energy

CTF084

Extensive and Intensive properties

3. Which of the following physical quantities are both independent from amount and path?

(A) Internal Energy
(B) Temperature
(C) Molar heat exchanged at constant volume
(D) Molar heat exchanged at constant pressure

Rankers Academy JEE

CTF082

4. Intensive properties are

(A) Density
(B) Entropy
(C) Boiling point
(D) Heat capacity

CTF083

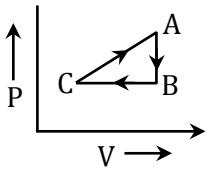
5. Among the following, the intensive property is (properties are) :

(A) Molar conductivity
(B) E.M.F of a cell
(C) Resistance
(D) Heat capacity

CTF085

Internal Energy

6. 1 mole of monoatomic ideal gas perform the following operation :



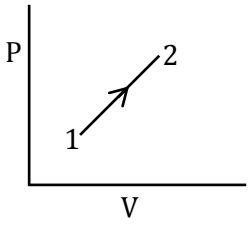
Select correct statements :

(A) A to B is endothermic process.
(B) B to C is endothermic process.
(C) C to A is endothermic process.
(D) ΔU in cyclic process ABCA is zero.

CTF096

For More Join: @IITJEE_Advanced

7. Determine the sign of ΔU, w & q in the following process :



(A) $\Delta U > 0$ (B) $q > 0$

(C) $w < 0$ (D) $w > 0$

CTF102

Degree of freedom

8. For a diatomic gas which options is/are correct :

(A) $\gamma = 1.40$ (B) $C_p = \frac{7R}{2}$

(C) $C_v = \frac{5R}{2}$ (D) $\gamma = 1.67$

CTF114

Isochoric process

9. Ten moles of an ideal gas ($\gamma = 1.2$) is heated from 27°C to 47°C at constant volume. Which of the following is incorrect :

Rankers Academy JEE

(A) $w > 0$ (B) $w = 0$

(C) $\Delta U = 200R$ (D) $\Delta H = 240R$

CTF125

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Extensive and Intensive properties

1. How many of the following are intensive

(a) Temperature (b) Internal Energy (c) Heat

(d) Molar volume (e) Molar heat capacity (f) viscosity

CTF086

Work Done

2. Water expands when it freezes. Determine amount of work (in joules), done when a system consisting of 1.0 L of liquid water freezes under a constant pressure of 1.0 atm and forms 1.1 L of ice.

CTF026

First Law of Thermodynamics

Rankers Academy JEE

3. The gas is cooled and loses 65 J of heat. The gas contracts as it cools and work done on the system equal to 20 J which is exchanged with the surroundings. What is the value of ΔE (in joules)?

CTF022

Degree of freedom

4. What is the degree of freedom of NH_3 at room temperature ?

CTF106

5. Find the value of $\left(\frac{X}{Y}\right)\times Z$

Where X = Total degree of freedom of O_2

Y = Total degree of freedom of Ne

Z = Total degree of freedom of H_2O

(Consider vibrational degree of freedom)

CTF030

Isothermal Process

6. Five moles of an ideal gas at 300K, expanded isothermally from an initial pressure of 4 atm to a final pressure of 1 atm against a constant external pressure of 1 atm. Calculate the corresponding value of the total work done (in kJ) all if expansion process is followed by reversible compression process for same value of pressure and temperature (ln2 = 0.7, R = 8.3 J/kmol)

CTF028

For More Join: @IITJEE_Advanced

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Work Done

7. Find the work done (in atm-litre) when 1 mole of gas expands from 1 litre to 5 litre against constant of 1 atm pressure.

CTF127

Heat and Heat Capacity (Cm, Cs, Cp, Cv)

8. A gas expands from 3 dm^3 to 5 dm^3 against a constant pressure of 3 atm. The work done during expansion is used to heat 10 mol of water of temperature 290 K. Calculate final temperature of water. Specific heat of water = 4.184 J g^- K.

CTF116

Enthalpy

9. Calculate the time (min.) for which a 100 watt electric heater must operate to evaporate 50 g sample of benzene at its boiling point. Given, standard enthalpy of vaporisation of benzene equal to 30.8 kJ/mol
(power = energy/time; 1 W = 1 J s^{-1}) **CTF103**

10. Enthalpy change in calorie for 1 mol of diatomic ideal gas under going the state change (2 atm, 300 K) to (3 atm, 360 K) will be ?

CTF145

Rankers Academy JEE

First Law of Thermodynamics

11. 1 mol of an ideal diatomic gas is expanded by heating from state-I (10 litre, 500K) to (20 litre, 800K) against constant pressure 2 bar. Find heat supplied in joule.
(Given R = 25/3 J/mole/K)

CTF128

12. When 1 mole of ice melt at 0°C and at constant pressure of 1 atm, 1440 calories of heat are absorbed by the system. The molar volumes of ice and water are 0.0196 and 0.0180 litre respectively. Calculate ΔH and ΔU (in calori) for the reaction.

CTF137

13. In an insulated container 1 mol of a liquid, molar volume 100 mL is at 1 bar. Liquid steeply taken to 100 bar, when volume of liquid decreases by 1 mL. Find ΔH and ΔU for the process in terms of bar–mL.

CTF115

Degree of freedom

14. Find total degree of freedom and break up as translational, rotational or vibrational degree of freedom in following cases.
(i) CO_2 (ii) SO_2 (iii) He (iv) NH_3

CTF142

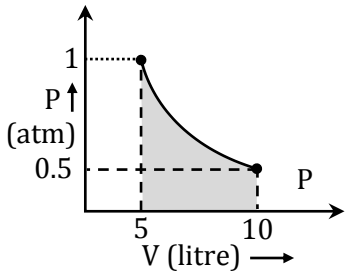
Isothermal Process

15. At 3 atmosphere pressure and 273 K, 1.5 L of H_2 gas allowed to expand isothermally at constant pressure of one atmosphere till pressure of the gas became one atmosphere. Calculate the work of expansion of H_2 gas.

CTF105

For More Join: @IITJEE_Advanced

16. Shaded area in the graph will be (in atm-L) : (ln2 = 0.7)



CTF129

17. An ideal gas expands isothermally from 10^{-3} m^3 to 10^{-2} m^3 at 300 K against a constant pressure of 10^5 Nm^{-2}. Calculate work in joule.

CTF130

18. An ideal gas undergoes expansion from A (10 atm, 1 litre) to B (1 atm, 10 litre), first against 5 atm and then against 1 atm, isothermally. Calculate the amount of heat absorbed (in litre-atm)

CTF131

19. Find the work done (in kJ) when one mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 27°C. (ln5 = 1.6)

CTF138

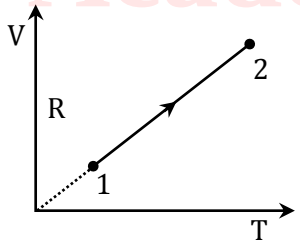
Isobaric process

20. Calculate the work done (in joule) by a system when one mol of a gas occupying 2 L volume is allowed to expand to 10 L volume against an external pressure of one atmosphere.

CTF104

21. Volume vs Temperature graph of 3 moles of O_2 gas is as shown in figure. The ratio of heat absorbed & the work done by the gas in process 1-2 is -

Rankers Academy JEE



CTF136

Adiabatic process

22. One mole of an ideal monoatomic gas undergoes expansion along a straight line on P–V curve from initial state A(3L, 8 atm) to final state B(7.5 L, 2 atm). Calculate q for the above process in L atm

CTF132

23. A gas is expanded in adiabatic process from 2L to 5L against constant external pressure of 1 bar then internal energy change in bar litre will be :

CTF133

24. Calculate the work done in calories by system in an irreversible (single step) adiabatic expansion of 1 mole of a polyatomic gas (γ = 4/3) from 400 K and pressure 16 atm to 1 atm. Also calculate work done in kcal if process done reversibly and adiabatically.

CTF134

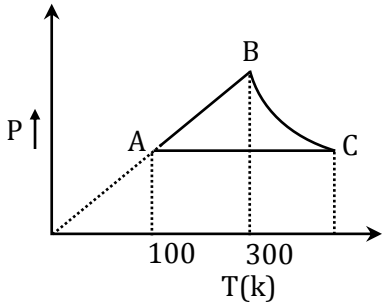
25. A monoatomic ideal gas undergoing irreversible adiabatic compression from 4L to 1L against 1 bar pressure. Calculate the enthalpy change of gas (in J) . (Given: 1 bar-L = 100 J)

CTF135

For More Join: @IITJEE_Advanced

Polytropic process

26. Calculate the net work done in the following cycle for one mol of an ideal gas (in calorie), where in process BC, PT = constant. (R = 2cal/mol-K).



CTF139

27. $\frac{1}{22.4}$ mol of an ideal monoatomic gas undergoes a reversible process for which $PV^2 = C$. The gas is expanded from initial volume of 1 L to final volume of 2 L starting from initial temperature of 273 K. Find the heat exchanged q during the process. Express your answer in litre atm.

$R = \frac{22.4}{273}$ L atm mole^{-1} degree^{-1}.

CTF140

28. For 1 mole of ideal monoatomic gas if $\frac{P}{V^2}$ = constant and initial temperature is 100 K. If gas is expanded from 1 L to 2 L then find

(a) heat capacity of the process (in cal/K-mol)

(b) total heat absorbed (in cal)

(c) work (in cal)

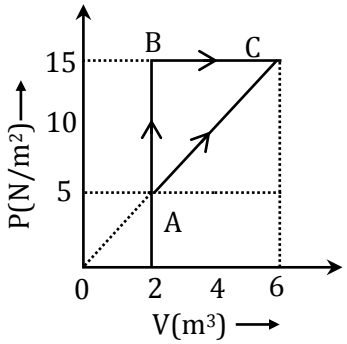
(d) (in cal) change in internal energy.

(Assume R = 0.0821 $\frac{L-atm}{mole-k}$, 1 L-atm = 24 cal)

Bankers Academy JEE

CTF141

29. The given figure shows a change of state A to state C by two paths ABC and AC for an ideal gas. Calculate the :



(a) Internal energy at C if the internal energy (in Joule) of gas at A is 10 J and amount of heat supplied to change its state to C through the path AC is 200J.

CTF143

(b) Amount of heat (in joule) supplied to the gas to go from A to B, if internal energy change of gas from A to B is 10 J.

CTF144

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. What happens when methane undergoes combustion in systems A and B respectively?

Adiabatic system	Diathermic container
System A	System B

[JEE (Main) 2023]

(1)

System A	System B
Temperature rises	Temperature remains same

(2)

System A	System B
Temperature falls	Temperature rises

(3)

System A	System B
Temperature falls	Temperature remains same

(4)

System A	System B
Temperature remains same	Temperature rises

Rankers Academy JEE

CTF147

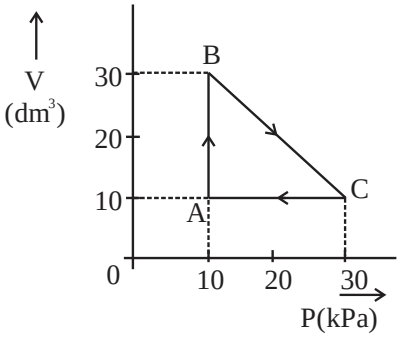
2. The total number of intensive properties from the following is ____

Volume, Molar heat capacity, Molarity, E^{θ}_{cell}, Gibbs free energy change, Molar mass, Mole

[JEE (Main) 2023]

CTF148

3.



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A → B → C → A as shown in the diagram. The total work done in the process is ____ J.

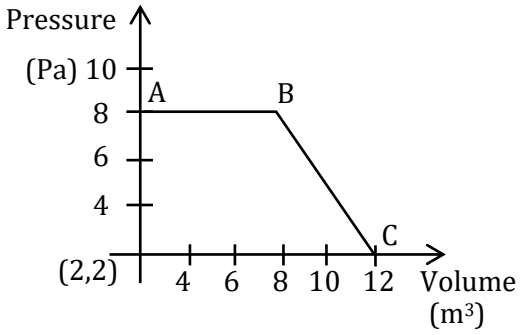
[JEE (Main) 2024]

CTF165

For More Join: @IITJEE_Advanced

4. The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is _____

[JEE (Main) 2020]



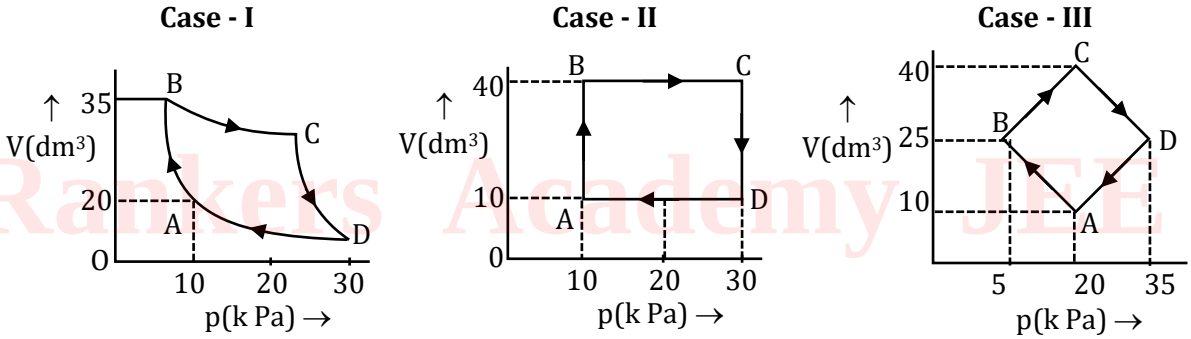
CTF046

5. When a 60 W electric heater is immersed in a gas for 100s in a constant volume container with adiabatic walls, the temperature of the gas rises by 5°C. The heat capacity of the given gas is _____ J K^{-1} (Nearest integer)

[JEE (Main) 2023]

CTF149

6.



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A→B→C→D→A as shown in the three cases above.

Choose the *correct* option regarding ΔU.

[JEE (Main) 2025]

(1) ΔU (Case-III) > ΔU (Case-II) > ΔU (Case-I)

(2) ΔU (Case-I) > ΔU (Case-II) > ΔU (Case-III)

(3) ΔU (Case-I) > ΔU (Case-III) > ΔU (Case-II)

(4) ΔU (Case-I) = ΔU (Case-II) = ΔU (Case-III)

CTF196

7. The molar heat capacity for an ideal gas at constant pressure is 20.785 J $K^{-1}mol^{-1}$. The change in internal energy is 5000 J upon heating it from 300K to 500K. The number of moles of the gas at constant volume is _____ [Nearest integer]

(Given: R = 8.314 J K^{-1} mol^{-1})

[JEE (Main) 2022]

CTF146

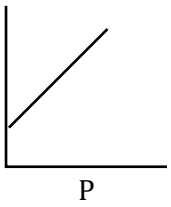
8. At constant volume, 4 mol of an ideal gas when heated from 300 K to 500K changes its internal energy by 5000 J. The molar heat capacity at constant volume is _____.

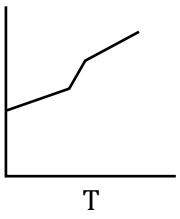
[JEE (Main) 2020]

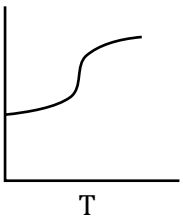
CTF047

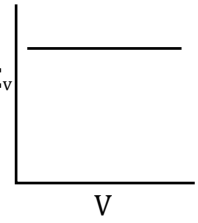
For More Join: @IITJEE_Advanced

9. For a diatomic ideal gas in a closed system, which of the following plots does not correctly describe the relation between various thermodynamic quantities ? **[JEE (Main) 2019]**

(1) 

(2) 

(3) 

(4) 

CTF043

10. If three moles of an ideal gas at 300 K expand isothermally from 30 dm^3 to 45 dm^3 against a constant opposing pressure of 80 kPa, then the amount of heat transferred is_____ J. **[JEE (Main) 2024]**

CTF166

11. During compression of a spring the work done is 10 kJ and 2 kJ escaped to the surroundings as heat. The change in internal energy, "U" (in kJ) is : **[JEE (Main) 2019]**

(1) –12 (2) –8 (3) 8 (4) 12

CTF040

12. Which one of the following equations does not correctly represent the first law of thermodynamics for the given process involving an ideal gas ? (Assume non expansion work is zero) **[JEE (Main) 2019]**

(1) Cyclic process : q = –w
(2) Adiabatic process : $\Delta U = -w$
(3) Isochoric process : $\Delta U = q$
(4) Isothermal process : q = –w

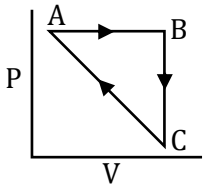
CTF041

13. 5 mole of an ideal gas at 100 K are allowed to undergo reversible compression till its temperature becomes 200 K. If C_v= 28 J K^{-1} mol^{-1}, calculate ΔU and ΔPV for this process. (R = 8.0 j K^{-1} mol^{-1}) **[JEE (Main) 2019]**

(1) ΔU = 14 kJ; Δ(PV) = 18 kJ
(2) ΔU = 14 kJ; Δ(PV) = 0.8 kJ
(3) ΔU = 14 kJ; Δ(PV) = 4 kJ
(4) ΔU = 14 kJ; Δ(PV) = 8.0 kJ

CTF042

14. An ideal gas undergoes a cyclic process as shown in figure.



Given :
ΔU_{BC} = –5 kJ mol^{-1}, q_{AB} = 2 kJ mol^{-1}
w_{AB} = –5 kJ mol^{-1}, W_{CA} = 3 kJ mol^{-1}
Heat absorbed by the system during process CA is :- **[JEE (Main) 2018]**

(1) 18 kJ mol^{-1} (2) +5 kJ mol^{-1} (3) –5 kJ mol^{-1} (4) –18 kJ mol^{-1}

CTF036

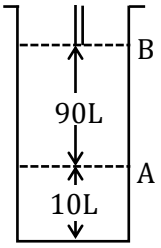
15. If 5 moles of an ideal gas expand from 10 L to a volume of 100 L at 300 K under isothermal and reversible condition then work, w, is –x J. The value of x is _____.
(Given $R = 8.314\ J\ K^{-1}mol^{-1}$)

[JEE (Main) 2024]

CTF167

For More Join: @IITJEE_Advanced

16.



Consider the figure provided.

1 mol of an ideal gas is kept in a cylinder, fitted with a piston, at the position A, at 18°C. If the piston is moved to position B, keeping the temperature unchanged, then 'x' L atm work is done in this reversible process.

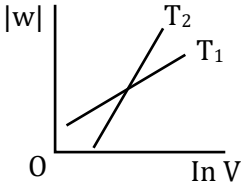
x = _____ L atm. (nearest integer)

[Given : Absolute temperature = °C + 273.15, R = 0.08206 L atm mol^{-1} K^{-1}] **[JEE (Main) 2024]**

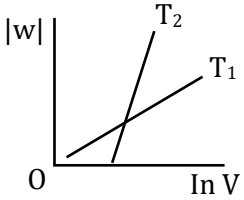
CTF168

17. Consider the reversible isothermal expansion of an ideal gas in a closed system at two different temperatures T_1 and T_2 ($T_1 < T_2$). The correct graphical depiction of the dependence of work done (w) on the final volume (V) is: **[JEE (Main) 2019]**

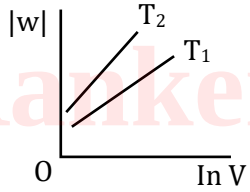
(1)



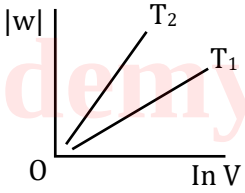
(2)



(3)



(4)



Rankers Academy JEE

CTF037

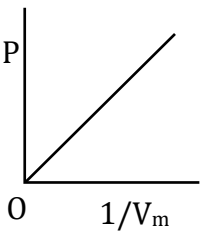
18. An ideal gas undergoes isothermal compression from 5 m^3 to 1 m^3 against a constant external pressure of 4 Nm^{-2}. Heat released in this process is used to increase the temperature of 1 mole of Al. If molar heat capacity of Al is 24 J mol^{-1} K^{-1}, the temperature of Al increases by **[JEE (Main) 2019]**

(1) $\frac{3}{2}$ K (2) $\frac{2}{3}$ K (3) 1 K (4) 2 K

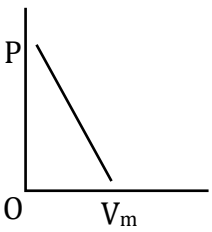
CTF038

19. The combination of plots which does not represent isothermal expansion of an ideal gas is : **[JEE (Main) 2019]**

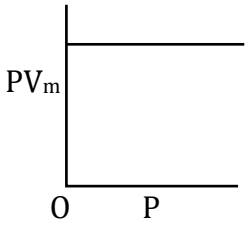
(A)



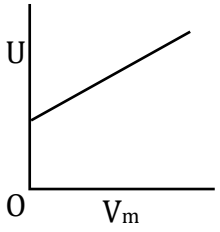
(B)



(C)



(D)



(1) (B) and (C) (2) (A) and (D) (3) (B) and (D) (4) (A) and (B)

CTF044

20. Three moles of an ideal gas are compressed isothermally from 60 L to 20 L using constant pressure of 5 atm. Heat exchange Q for the compression is – ___ Lit. atm. **[JEE (Main) 2024]**

CTF169

For More Join: @IITJEE_Advanced

21. An ideal gas is allowed to expand from 1L to 10L against a constant external pressure of 1 bar. The work done in kJ is :

[JEE (Main) 2019]

(1) –9.0 (2) +10.0 (3) –0.9 (4) –2.0

CTF039

22. Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following: **[JEE (Main) 2024]**

(1) $q = 0, \Delta T \neq 0, w = 0$ (2) $q = 0, \Delta T < 0, w \neq 0$

(3) $q \neq 0, \Delta T = 0, w = 0$ (4) $q = 0, \Delta T = 0, w = 0$

CTF170

23. Five moles of an ideal gas at 1 bar and 298 K is expanded into vacuum to double the volume. The work done is _____ J. **[JEE (Main) 2020]**

CTF045

24. A liquid when kept inside a thermally insulated closed vessel at 25°C was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters ?

[JEE (Main) 2025]

(1) $\Delta U > 0, q = 0, w > 0$ (2) $\Delta U = 0, q = 0, w = 0$

(3) $\Delta U < 0, q = 0, w > 0$ (4) $\Delta U = 0, q < 0, w > 0$

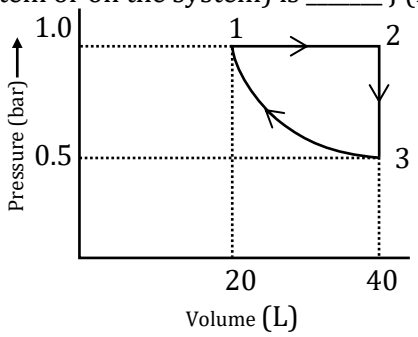
CTF197

25. An ideal gas, $\bar{C}_V = \frac{5}{2}R$, is expanded adiabatically against a constant pressure of 1 atm until it doubles in volume. If the initial temperature and pressure is 298 K and 5 atm, respectively then the final temperature is _____ K (nearest integer).

[$\bar{C}_V$ is the molar heat capacity at constant volume] **[JEE (Main) 2024]**

CTF171

26. One mole of an ideal monoatomic gas is subjected to changes as shown in the graph. The magnitude of the work done (by the system or on the system) is _____ J (nearest integer).



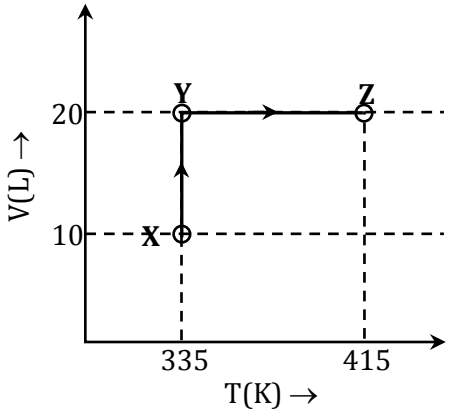
Given : log 2 = 0.3, ln 10 = 2.3 **[JEE (Main) 2023]**

CTF150

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Consider the following volume-temperature (V – T) diagram for the expansion of 5 moles of an ideal monoatomic gas.



Considering only P-V work is involved, the total change in enthalpy (in Joule) for the transformation of state in the sequence **X → Y → Z** is _____ .

[Use the given data: Molar heat capacity of the gas for the given temperature range, $C_{V,m}$ = 12 J K^{-1} mol^{-1} and gas constant, R = 8.3 J K^{-1} mol^{-1}]

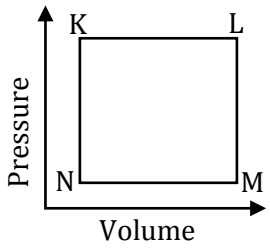
[JEE (Advanced) 2024]

CTF172

Rankers Academy JEE

Paragraph for Question # 2 and 3

A fixed mass 'm' of a gas is subjected to transformation of states from K to L to M to N and back to K as shown in the figure.



2. The pair of isochoric processes among the transformation of states is

[JEE (Advanced) 2013]

(A) K to L and L to M
(B) L to M and N to K
(C) L to M and M to N
(D) M to N and N to K

CTF071

3. The succeeding operations that enable this transformation of states are

[JEE (Advanced) 2013]

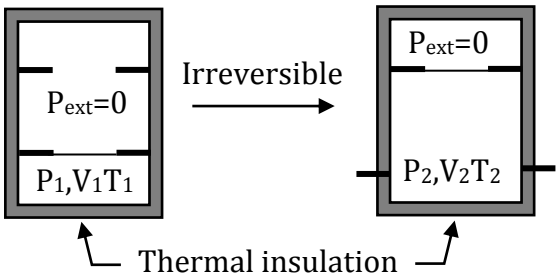
(A) Heating, cooling, heating, cooling
(B) Cooling, heating, cooling, heating
(C) Heating, cooling, cooling, heating
(D) Cooling, heating, heating, cooling

CTF072

For More Join: @IITJEE_Advanced

4. An ideal gas in thermally insulated vessel at internal pressure = P_1, volume = V_1 and absolute temperature = T_1 expands irreversibly against zero external pressure, as shown in the diagram. The final internal pressure, volume and absolute temperature of the gas are P_2, V_2 and T_2, respectively. For this expansion,

[JEE (Advanced) 2014]

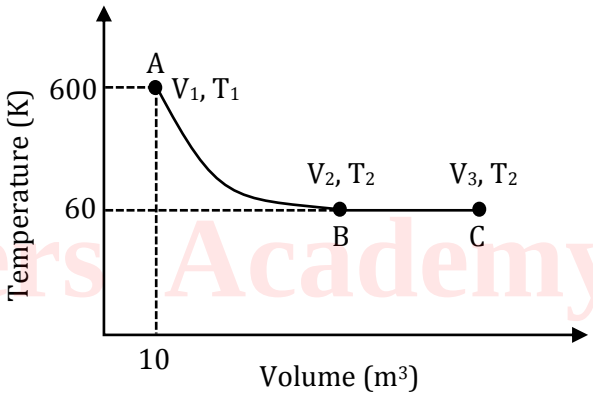


(A) $q = 0$ (B) $T_2 = T_1$

(C) $P_2V_2 = P_1V_1$ (D) $P_2V_2^{\gamma} = P_1V_1^{\gamma}$

CTF073

5. One mole of an ideal monoatomic gas undergoes two reversible processes (A → B and B → C) as shown in the given figure :



Rankers Academy JEE

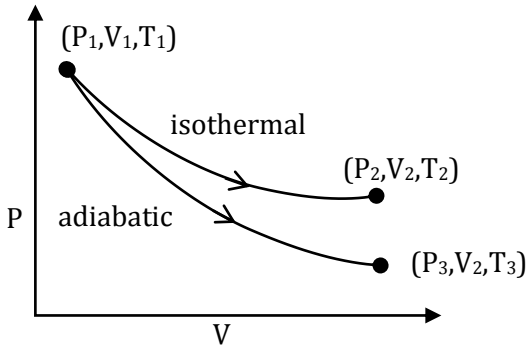
A → B is an adiabatic process. If the total heat absorbed in the entire process (A → B and B → C) is $RT_2 \ln 10$, the value of $2 \log V_3$ is _____.

[Use, molar heat capacity of the gas at constant pressure, $C_{p,m} = \frac{5}{2}R$]

[JEE (Advanced) 2023]

CTF151

6. The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure. Which of the following statement(s) is (are) correct? **[JEE (Advanced) 2012]**

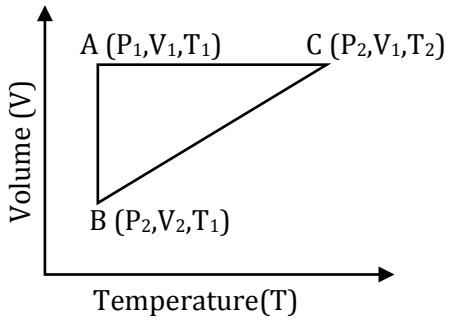


(A) $T_1 = T_2$ (B) $T_3 > T_1$ (C) $w_{isothermal} > w_{adiabatic}$ (D) $\Delta U_{isothermal} > \Delta U_{adiabatic}$

CTF070

For More Join: @IITJEE_Advanced

7. A reversible cyclic process for an ideal gas is shown below. Here, P, V and T are pressure, volume and temperature, respectively. The thermodynamic parameters q, w, H and U are heat, work, enthalpy and internal energy, respectively. **[JEE (Advanced) 2018]**



The correct option(s) is (are)

(A) $q_{AC} = \Delta U_{BC}$ and $w_{AB} = P_2 (V_2 - V_1)$

(B) $w_{BC} = P_2 (V_2 - V_1)$ and $q_{BC} = \Delta H_{AC}$

(C) $\Delta H_{CA} < \Delta U_{CA}$ and $q_{AC} = \Delta U_{BC}$

(D) $q_{BC} = \Delta H_{AC}$ and $\Delta H_{CA} > \Delta U_{CA}$

CTF075

8. An ideal gas is expanded from (p_1, V_1, T_1) to (p_2, V_2, T_2) under different conditions. The correct statement(s) among the following is(are) :- **[JEE (Advanced) 2017]**

(A) The work done on the gas is maximum when it is compressed irreversibly from (p_2, V_2) to (p_1, V_1) against constant pressure p_1

(B) The work done by the gas is less when it is expanded reversibly from V_1 to V_2 under adiabatic conditions as compared to that when expanded reversibly from V_1 to V_2 under isothermal conditions.

(C) The change in internal energy of the gas (i) zero, if it is expanded reversibly with $T_1 = T_2$, and (ii) positive, if it is expanded reversibly under adiabatic conditions with $T_1 \neq T_2$

(D) If the expansion is carried out freely, it is simultaneously both isothermal as well as adiabatic

CTF074

Rankers Academy JEE

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	B	D	A	A	C	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	D	A	C	A	C	A	D	B
Que.	21	22	23	24	25	26	27	28	29	
Ans.	A	D	A	D	C	C	A	C	A	

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	
Ans.	ABC	ACD	BCD	AC	AB	CD	ABC	ABC	ACD	

EXERCISE (S)

1. 4 **2.** –10 **3.** –45 **4.** 6 **5.** 18

6. 8 **7.** –4.00 **8.** 290.8 **9.** 3.29 **10.** 420.00

11. 8250.00 **12.** 1440.00 (1440.03 to 1440.04) **13.** 9900.00

14. (i) 9, 3, 2, 4 (ii) 9, 3, 3, 3 (iii) 3, 3, 0, 0 (iv) 12, 3, 3, 6 **15.** 303.9 **16.** 3.50

17. –900.00 **18.** 13.00 **19.** –3.90 to –4.00 **20.** –810.4 **21.** 3.50

22. 9.00 **23.** –3.00 **24.** –562.50, –1.20 **25.** 500.00

26. –800.00 **27.** –0.25 **28.** (a) 3.67 ; (b) 2566.67 ; (c) –466.67 ; (d) 2100.00)

29. (a) 170.00 (b) 10.00

Rankers Academy JEE

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	200	48	1200	4	2	6.25	1	1200
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	3	2	28721	55	2	2	3	200
Que.	21	22	23	24	25	26				
Ans.	3	4	0	1	274	620				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	
Ans.	8120	B	C	ABC	7	ACD	BC	ABD	

For More Join: @IITJEE_Advanced